Supplementary information for constrained annealing verification

Numerical verification and feasibility supplement · 1 November 2026

This supplement documents genuine reference calculations and published material data. No production simulation, independent validation prediction, sensitivity index, uncertainty interval, Pareto solution or optimization confirmation exists. Missing results are not zero responses. Tables S1–S12 retain the earlier reference evidence. Their numeric citations use the legacy reference key in Section M4 below, not the shorter main article bibliography. Sections M1–M3 document the new investigation. Statements about production incompleteness remain applicable; non-isothermal reference verification has now been added.

S1. Literature comparison and reproducibility tables

Literature values below are source evidence, not new experiments or solver findings. Full matrices, formulation flags and exact locators remain in the repository. Reference identities were checked against the bibliography. The focused September 2026 audit adds an ABS clearance precedent [41] without treating it as PLA material evidence.

Table S1. Closest competing studies and implications for scope

Study	Established overlap	Distinction still requiring evidence
Bute et al. [5]	Irreversible strain and ANSYS printing stress	Predictive annealing-clearance/contact relation
Chapuis et al. [8]	Viscoelastic pre-strain recovery FE	Fixture-controlled dimensional preservation
Issabayeva and Shishkovsky [9]	ANSYS, DMA, Prony and thermal response	Released fixture geometry and stress
Wijnen et al. [6]	Warpage model and annealing dimensions	Quantified annealing restraint
Trofimov et al. [7]	Thermal/deformation prediction and comparison	Post-print annealing contact cycle
Lluch-Cerezo et al. [11]	Powder-supported dimensional control	Explicit clearance and contact mechanics
Wijnbergen et al. [12]	Tough-PLA annealing media	Compatible grade and plate-gap prediction
Chiscop et al. [24]	Encapsulation across geometries	Contact-clearance response under release
De Assis et al. [13]	Geometry/mechanics and thermal conditioning	Transient fixture-contact model
Ben Amor and Souissi [14]	Annealed PLA, FE and optimization	Released dimensions–warpage–stress decision

Table S2. Prusament PLA generalized-Maxwell coefficients at T_g = 65 °C from Chapuis et al. [8, supplementary Table B.2]

i	k_i (MPa)	τ_i (s)
1	6.228	1.176×10^{-14}
2	1.806	1.228×10^{-14}
3	7.849	2.878×10^{-14}
4	12.751	1.701×10^{-12}
5	20.857	9.178×10^{-12}
6	31.104	6.034×10^{-11}
7	45.566	6.187×10^{-9}
8	64.036	1.102×10^{-7}
9	92.484	1.047×10^{-6}
10	135.916	6.903×10^{-6}
11	113.488	2.582×10^{-5}
12	128.266	5.363×10^{-5}
13	173.470	1.838×10^{-4}
14	93.393	4.386×10^{-4}
15	163.804	7.577×10^{-4}
16	134.311	1.795×10^{-3}
17	119.266	2.292×10^{-3}
18	100.596	6.576×10^{-3}
19	80.534	9.920×10^{-3}
20	61.624	3.290×10^{-2}
21	44.508	5.374×10^{-2}
22	30.529	6.586×10^{-2}
23	18.610	3.503×10^{1}

Table S3. Candidate AISI 304 fixture properties from Meng et al. [38, Table 1]

Property	20 °C	100 °C / 200 °C
ρ	7910 kg m^{-3}	7876 / 7840 kg m^{-3}
c_p	456 J kg^{-1} K^{-1}	494 / 532 J kg^{-1} K^{-1}
k	16.2 W m^{-1} K^{-1}	16.6 / 17.45 W m^{-1} K^{-1}
α	15.5 μm m^{-1} K^{-1}	16.3 / 16.7 μm m^{-1} K^{-1}
E	200 GPa	191.4 / 183.5 GPa
v	0.290	0.285 / 0.289

Table S4. Reserved experimental dimensional means from Lluch-Cerezo et al. [11, Table 5], without mould

T (°C)	Length change (%)	Width change (%)	Height change (%)
63	−0.13	−0.06	0.00
75	−1.60	−0.11	2.74
86	−2.30	−0.16	2.62
98	−2.88	0.15	2.60
109	−3.05	0.30	3.94
132	−3.58	0.18	3.69

Table S5. Genuine ANSYS–analytical center-temperature comparison for the plane-wall verification

Time (s)	ANSYS / analytical (°C)	Absolute / excursion-relative error
150	21.892109254 / 21.889130841	0.002978413 °C / 0.157660518%
300	27.631062216 / 27.625947421	0.005114795 °C / 0.067070946%
450	34.805348857 / 34.801718199	0.003630658 °C / 0.024528625%
600	41.003870402 / 41.001387994	0.002482407 °C / 0.011820207%
900	50.967098637 / 50.966218772	0.000879865 °C / 0.002841371%
1050	53.056900023 / 53.059537028	0.002637005 °C / 0.007976533%
1200	50.753732030 / 50.758912731	0.005180700 °C / 0.016842924%
1500	42.903452036 / 42.906520281	0.003068244 °C / 0.013394633%
1800	37.051786082 / 37.053501761	0.001715679 °C / 0.010060566%

Table S6. Isothermal Prony finite-ramp stress verification at 65 °C

Time (s)	ANSYS stress (MPa)	Analytical stress (MPa)	Absolute error (MPa)
1	0.0375536568	0.0375574835	0.0000038266
11	0.0243886374	0.0243886379	0.0000000005
31	0.0183896348	0.0183896352	0.0000000005
32	−0.0193833038	−0.0193871309	0.0000038271
62	−0.0045757745	−0.0045757747	0.0000000001

Table S7. Contact-control sensitivity on the 40-element interface

Control	Maximum penetration (mm)	Mean / peak pressure (MPa)
Penalty or augmented, F_{KN} = 0.1	1.12141×10^{-3}	16.9989 / 17.9426
Penalty or augmented, F_{KN} = 1	1.52677×10^{-4}	17.5368 / 24.4284
Penalty or augmented, F_{KN} = 10	1.80923×10^{-5}	17.6242 / 28.9476
Normal Lagrange, μ = 0	0	17.5721 / 21.3788
Augmented, F_{KN} = 1, μ = 0.1	1.97427×10^{-4}	18.4343 / 31.5883
Augmented, F_{KN} = 1, μ = 0.3	3.24243×10^{-4}	20.0334 / 51.8788

S2. Evidence and reproduction boundaries

The original literature matrix contains 34 journal studies; the separate audit addendum records the additional ABS clearance study [41]. The base paper is immutable. Property records preserve units, temperatures, formulation, exact source location and use restrictions. Full convergence CSVs contain 39 mesh, 24 time-step and nine contact-control rows; these are response records, not independent experiments. All 43 technical convergence directories remain archived, including 15 excluded attempts.

Verification constants, load schedules, relaxation mismatch and clearance screens are design choices. The assembly construction is not a solved annealing model. The validation archive retains 45 published observations: 18 reserved means, 24 context records and three quarantined maxima. Prediction/error fields remain blank. Empty registries do not mean zero physical response.

Each run preserves input, script, command, raw output and extraction hashes. Corrections create new derived records or attempts; raw evidence is not overwritten. The supplementary files support numerical reproduction and do not supply missing physical validation.

S3. Full scalar verification and refinement records

The following tables reproduce every accepted scalar comparison and each audited mesh/time response record. Raw nodal and element exports, complete input decks, warnings and rejected attempts remain in the repository at the paths indexed by result_traceability.csv. Units follow each quantity; reaction is force in the structural verification and force per unit depth in plane-strain contact convergence. Percent errors at a zero reference are undefined, not zero.

Table S8. All structural reference comparisons

Case	t (s)	Quantity	ANSYS	Reference	Absolute error
free	1	ux (mm)	0.001	0.001	2.998903×10^{-16}
free	1	uy (mm)	0.0001	0.0001	2.999174×10^{-17}
free	1	uz (mm)	0.0001	0.0001	2.999174×10^{-17}
free	1	stress x (MPa)	1.15001×10^{-15}	0	1.15001×10^{-15}
free	1	reaction x (N)	-1.15001×10^{-15}	0	1.15001×10^{-15}
free	2	ux (mm)	0.006	0.006	8.396062×10^{-16}
free	2	uy (mm)	0.0006	0.0006	7.892992×10^{-17}
free	2	uz (mm)	0.0006	0.0006	7.795414×10^{-17}
free	2	stress x (MPa)	6.911789×10^{-15}	0	6.911789×10^{-15}
free	2	reaction x (N)	$-6.911789 \times 10^{-15}$	0	6.911789×10^{-15}
free	3	ux (mm)	0	0	0
free	3	uy (mm)	0	0	0
free	3	uz (mm)	0	0	0
free	3	stress x (MPa)	0	0	0
free	3	reaction x (N)	0	0	0
fixed	1	ux (mm)	0	0	0
fixed	1	uy (mm)	0.00013	0.00013	3.897707×10^{-17}

Case	t (s)	Quantity	ANSYS	Reference	Absolute error
fixed	1	uz (mm)	0.00013	0.00013	3.897707×10^{-17}
fixed	1	stress x (MPa)	−0.2	−0.2	2.980232×10^{-9}
fixed	1	reaction x (N)	0.2	0.2	5.99798×10^{-14}
fixed	2	ux (mm)	0	0	0
fixed	2	uy (mm)	0.00078	0.00078	1.049508×10^{-16}
fixed	2	uz (mm)	0.00078	0.00078	1.038666×10^{-16}
fixed	2	stress x (MPa)	−1.2	−1.2	4.768372×10^{-8}
fixed	2	reaction x (N)	1.2	1.2	1.598721×10^{-13}
fixed	3	ux (mm)	0	0	0
fixed	3	uy (mm)	0	0	0
fixed	3	uz (mm)	0	0	0
fixed	3	stress x (MPa)	0	0	0
fixed	3	reaction x (N)	0	0	0
eigen free	1	ux (mm)	−0.009	−0.009	2.688821×10^{-16}
eigen free	1	uy (mm)	−0.0009	−0.0009	3.198396×10^{-17}
eigen free	1	uz (mm)	−0.0009	−0.0009	2.992398×10^{-17}
eigen free	1	stress x (MPa)	$-7.882584 \times 10^{-15}$	0	7.882584×10^{-15}
eigen free	1	reaction x (N)	7.882583×10^{-15}	0	7.882583×10^{-15}
eigen free	2	ux (mm)	−0.004	−0.004	7.901665×10^{-16}
eigen free	2	uy (mm)	−0.0004	−0.0004	7.995991×10^{-17}
eigen free	2	uz (mm)	−0.0004	−0.0004	7.697835×10^{-17}

Case	t (s)	Quantity	ANSYS	Reference	Absolute error
eigen free	2	stress x (MPa)	-3.108625 × 10^{-15}	-1.084202 × 10^{-16}	3.000204 × 10^{-15}
eigen free	2	reaction x (N)	3.108624 × 10^{-15}	1.084202 × 10^{-16}	3.000204 × 10^{-15}
eigen free	3	ux (mm)	−0.01	−0.01	1.00614 × 10^{-16}
eigen free	3	uy (mm)	−0.001	−0.001	1.951564 × 10^{-18}
eigen free	3	uz (mm)	−0.001	−0.001	0
eigen free	3	stress x (MPa)	-1.032507 × 10^{-14}	0	1.032507 × 10^{-14}
eigen free	3	reaction x (N)	1.032507 × 10^{-14}	0	1.032507 × 10^{-14}
eigen fixed	1	ux (mm)	0	0	0
eigen fixed	1	uy (mm)	−0.00117	−0.00117	4.011548 × 10^{-17}
eigen fixed	1	uz (mm)	−0.00117	−0.00117	4.011548 × 10^{-17}
eigen fixed	1	stress x (MPa)	1.8	1.8	4.768372 × 10^{-8}
eigen fixed	1	reaction x (N)	−1.8	−1.8	5.995204 × 10^{-14}
eigen fixed	2	ux (mm)	0	0	0
eigen fixed	2	uy (mm)	−0.00052	−0.00052	1.03975 × 10^{-16}
eigen fixed	2	uz (mm)	−0.00052	−0.00052	9.996344 × 10^{-17}
eigen fixed	2	stress x (MPa)	0.8	0.8	1.192093 × 10^{-8}
eigen fixed	2	reaction x (N)	−0.8	−0.8	1.609823 × 10^{-13}
eigen fixed	3	ux (mm)	0	0	0
eigen fixed	3	uy (mm)	−0.0013	−0.0013	0
eigen fixed	3	uz (mm)	−0.0013	−0.0013	0
eigen fixed	3	stress x (MPa)	2	2	0
eigen fixed	3	reaction x (N)	−2	−2	0

Case	t (s)	Quantity	ANSYS	Reference	Absolute error
eigen fixed	4	ux (mm)	−0.01	−0.01	1.00614×10^{-16}
eigen fixed	4	uy (mm)	−0.001	−0.001	1.951564×10^{-18}
eigen fixed	4	uz (mm)	−0.001	−0.001	0
eigen fixed	4	stress x (MPa)	$-1.032507 \times 10^{-14}$	0	1.032507×10^{-14}
eigen fixed	4	reaction x (N)	1.032507×10^{-14}	0	1.032507×10^{-14}
visco	1	ux (mm)	0.01	0.01	9.020562×10^{-17}
visco	1	uy (mm)	−0.00035	−0.00035	4.011548×10^{-18}
visco	1	uz (mm)	−0.00035	−0.00035	4.011548×10^{-18}
visco	1	stress x (MPa)	0.03755366	0.03755748	3.826628×10^{-6}
visco	1	reaction x (N)	−0.03755366	−0.03755748	3.827983×10^{-6}
visco	11	ux (mm)	0.01	0.01	6.071532×10^{-17}
visco	11	uy (mm)	−0.00035	−0.00035	2.005774×10^{-18}
visco	11	uz (mm)	−0.00035	−0.00035	2.005774×10^{-18}
visco	11	stress x (MPa)	0.02438864	0.02438864	4.999321×10^{-10}
visco	11	reaction x (N)	−0.02438864	−0.02438864	4.555384×10^{-15}
visco	31	ux (mm)	0.01	0.01	0
visco	31	uy (mm)	−0.00035	−0.00035	0
visco	31	uz (mm)	−0.00035	−0.00035	9.75782×10^{-19}
visco	31	stress x (MPa)	0.01838963	0.01838964	4.545142×10^{-10}
visco	31	reaction x (N)	−0.01838964	−0.01838964	2.428613×10^{-15}

Case	t (s)	Quantity	ANSYS	Reference	Absolute error
visco	32	ux (mm)	2.330601×10^{-14}	0	2.330601×10^{-14}
visco	32	uy (mm)	-8.15734×10^{-16}	0	8.15734×10^{-16}
visco	32	uz (mm)	$-8.150692 \times 10^{-16}$	0	8.150692×10^{-16}
visco	32	stress x (MPa)	−0.0193833	−0.01938713	3.82709×10^{-6}
visco	32	reaction x (N)	0.0193833	0.01938713	3.827983×10^{-6}
visco	62	ux (mm)	0	0	0
visco	62	uy (mm)	$-3.187388 \times 10^{-19}$	0	3.187388×10^{-19}
visco	62	uz (mm)	6.912625×10^{-19}	0	6.912625×10^{-19}
visco	62	stress x (MPa)	−0.004575775	−0.004575775	1.14099×10^{-10}
visco	62	reaction x (N)	0.004575775	0.004575775	1.56819×10^{-15}

Table S9. All gap-contact reference comparisons

Case	t (s)	Quantity	ANSYS	Reference	Absolute error
contact	1	ux (mm)	0.001	0.001	2.998903×10^{-16}
contact	1	uy (mm)	0.0001	0.0001	2.999174×10^{-17}
contact	1	uz (mm)	0.0001	0.0001	2.999174×10^{-17}
contact	1	stress x (MPa)	$-2.801724 \times 10^{-17}$	0	2.801724×10^{-17}
contact	1	reaction x (N)	2.801724×10^{-17}	0	2.801724×10^{-17}
contact	1	C force (N)	0	0	0
contact	1	C closed	0	0	0
contact	1	gap (mm)	0.001	0.001	4.749745×10^{-11}
contact	2	ux (mm)	0.002	0.002	6.700369×10^{-16}

Case	t (s)	Quantity	ANSYS	Reference	Absolute error
contact	2	uy (mm)	0.00072	0.00072	8.391725×10^{-17}
contact	2	uz (mm)	0.00072	0.00072	8.391725×10^{-17}
contact	2	stress x (MPa)	−0.8	−0.8	1.192093×10^{-8}
contact	2	reaction x (N)	0.8	0.8	2.68674×10^{-14}
contact	2	C force (N)	−0.8	−0.8	1.192093×10^{-8}
contact	2	C closed	1	1	0
contact	2	gap (mm)	0	0	0
contact	3	ux (mm)	2.001532×10^{-16}	0	2.001532×10^{-16}
contact	3	uy (mm)	2.001784×10^{-17}	0	2.001784×10^{-17}
contact	3	uz (mm)	2.001356×10^{-17}	0	2.001356×10^{-17}
contact	3	stress x (MPa)	8.932265×10^{-18}	0	8.932265×10^{-18}
contact	3	reaction x (N)	$-8.932262 \times 10^{-18}$	0	8.932262×10^{-18}
contact	3	C force (N)	0	0	0
contact	3	C closed	0	0	0
contact	3	gap (mm)	0.002	0.002	9.49949×10^{-11}

Table S10. All audited mesh and time refinement responses

Study	Case / level	Quantity	Value	Unit	Change (%)
Mesh	S mesh coarse a03	W_{max}	0.008170221	mm	
Mesh	S mesh coarse a03	resid. disp.	0.05963922	mm	
Mesh	S mesh coarse a03	resid. stress p95	9.965×10^{-5}	MPa	
Mesh	S mesh medium a01	W_{max}	0.01125978	mm	27.43893
Mesh	S mesh medium a01	resid. disp.	0.07853777	mm	24.06301
Mesh	S mesh medium a01	resid. stress p95	0.000494075	MPa	79.831

Study	Case / level	Quantity	Value	Unit	Change (%)
Mesh	S mesh fine a01	W_{max}	0.01250177	mm	9.934494
Mesh	S mesh fine a01	resid. disp.	0.08534517	mm	7.976313
Mesh	S mesh fine a01	resid. stress p95	0.0007046092	MPa	29.87957
Mesh	S mesh extra fine a01	W_{max}	0.01279686	mm	2.305902
Mesh	S mesh extra fine a01	resid. disp.	0.08674431	mm	1.612952
Mesh	S mesh extra fine a01	resid. stress p95	0.0007082847	MPa	0.5189303
Mesh	S mesh ultra fine a01	W_{max}	0.01291733	mm	0.9326807
Mesh	S mesh ultra fine a01	resid. disp.	0.08725225	mm	0.5821426
Mesh	S mesh ultra fine a01	resid. stress p95	0.0007229355	MPa	2.026572
Mesh	T mesh coarse a01	T lag at 300 s	52.37903	°C	
Mesh	T mesh coarse a01	temp. grad. at 300 s	252.1411	$K\ m^{-1}$	
Mesh	T mesh medium a01	T lag at 300 s	52.37115	°C	0.01505384
Mesh	T mesh medium a01	temp. grad. at 300 s	252.384	$K\ m^{-1}$	0.09626966
Mesh	T mesh fine a01	T lag at 300 s	52.36917	°C	0.003767935
Mesh	T mesh fine a01	temp. grad. at 300 s	252.4447	$K\ m^{-1}$	0.02403675
Mesh	T mesh extra fine a01	T lag at 300 s	52.36868	°C	0.0009422631
Mesh	T mesh extra fine a01	temp. grad. at 300 s	252.4599	$K\ m^{-1}$	0.006007273
Mesh	C mesh lagrange coarse a03	mean C press.	16.98722	MPa	
Mesh	C mesh lagrange coarse a03	max. C press.	19.76897	MPa	
Mesh	C mesh lagrange coarse a03	max. penetration	0	mm	
Mesh	C mesh lagrange coarse a03	normal reaction	172.4908	$N\ mm^{-1}$	
Mesh	C mesh lagrange medium a03	mean C press.	17.44909	MPa	2.646943
Mesh	C mesh lagrange medium a03	max. C press.	20.98779	MPa	5.807288
Mesh	C mesh lagrange medium a03	max. penetration	0	mm	

Study	Case / level	Quantity	Value	Unit	Change (%)
Mesh	C mesh lagrange medium a03	normal reaction	177.2646	N mm^{-1}	2.69307
Mesh	C mesh lagrange fine a03	mean C press.	17.57206	MPa	0.6997847
Mesh	C mesh lagrange fine a03	max. C press.	21.37878	MPa	1.82884
Mesh	C mesh lagrange fine a03	max. penetration	0	mm	
Mesh	C mesh lagrange fine a03	normal reaction	178.5694	N mm^{-1}	0.7306797
Mesh	C mesh lagrange extra fine a03	mean C press.	17.5944	MPa	0.1269802
Mesh	C mesh lagrange extra fine a03	max. C press.	21.45629	MPa	0.3612721
Mesh	C mesh lagrange extra fine a03	max. penetration	0	mm	
Mesh	C mesh lagrange extra fine a03	normal reaction	178.8167	N mm^{-1}	0.1383057
Time	S time coarse a01	W_{max}	0.01248075	mm	
Time	S time coarse a01	resid. disp.	0.085201	mm	
Time	S time coarse a01	resid. stress p95	0.0007069894	MPa	
Time	S time medium a01	W_{max}	0.01249482	mm	0.1126111
Time	S time medium a01	resid. disp.	0.08529745	mm	0.1130754
Time	S time medium a01	resid. stress p95	0.0007080892	MPa	0.155314
Time	S mesh fine a01	W_{max}	0.01250177	mm	0.05561255
Time	S mesh fine a01	resid. disp.	0.08534517	mm	0.05591337
Time	S mesh fine a01	resid. stress p95	0.0007046092	MPa	0.4938889
Time	S time extra fine a01	W_{max}	0.01250216	mm	0.00309372
Time	S time extra fine a01	resid. disp.	0.08534791	mm	0.003212977
Time	S time extra fine a01	resid. stress p95	0.0007036497	MPa	0.136358
Time	T time coarse a01	T lag at 300 s	52.2638	°C	
Time	T time coarse a01	temp. grad. at 300 s	251.9462	K m^{-1}	

Study	Case / level	Quantity	Value	Unit	Change (%)
Time	T time coarse a01	temp. profile L2 excursion	23.72262	°C	
Time	T time medium a01	T lag at 300 s	52.35208	°C	0.1686272
Time	T time medium a01	temp. grad. at 300 s	252.3786	K m^{-1}	0.1713168
Time	T time medium a01	temp. profile L2 excursion	23.74377	°C	0.08906041
Time	T mesh extra fine a01	T lag at 300 s	52.36868	°C	0.03169506
Time	T mesh extra fine a01	temp. grad. at 300 s	252.4599	K m^{-1}	0.03220043
Time	T mesh extra fine a01	temp. profile L2 excursion	23.74776	°C	0.01683771
Time	T time extra fine a01	T lag at 300 s	52.37145	°C	0.005284926
Time	T time extra fine a01	temp. grad. at 300 s	252.4734	K m^{-1}	0.005369188
Time	T time extra fine a01	temp. profile L2 excursion	23.74843	°C	0.002810575

S4. Material property inventory and use restrictions

The property registry below preserves all recorded numerical PLA values, including comparator formulations that are not admitted to the reference material card. P = Prusament PLA; all other formulations are comparator evidence. Exact print settings, confidence, full source locators, temperature context and ANSYS-use restrictions are retained in material/pla_properties.csv. A number in this inventory is not authorization to mix formulations. Tables S2–S3 provide the reference spectrum and candidate fixture data.

Table S11. Complete PLA property inventory with source and formulation

ID / source	Property / component	Value / unit	Temperature / formulation
PLA-001 PrusamentTDS2022	ρ / isotropic	1240 kg m^{-3}	not reported Prusament PLA
PLA-002 PrusamentTDS2022	E / horizontal	2300 MPa	not reported Prusament PLA
PLA-003 PrusamentTDS2022	E / vertical xz	2400 MPa	not reported Prusament PLA
PLA-004 Chapuis2025	ν / isotropic	0.35	23–85 °C Prusament PLA
PLA-005 Chapuis2025	α / isotropic	68 µm m^{-1} K^{-1}	23–85 °C Prusament PLA
PLA-006 Chapuis2025	T_g / tan δ peak	65 °C	65 °C Prusament PLA

ID / source	Property / component	Value / unit	Temperature / formulation
PLA-007 Chapuis2025	E_∞ / equilibrium Hookean element k_0	10.598 MPa	65 °C Prusament PLA
PLA-008 Chapuis2025	E_0 / $k_0 + \Sigma k_i$	1691.594 MPa	65 °C Prusament PLA
PLA-009 Chapuis2025	C_1 / above T_g	17.4	≥65 °C Prusament PLA
PLA-010 Chapuis2025	C_2 / above T_g	51.6 K	≥65 °C Prusament PLA
PLA-011 Chapuis2025	C_3 / below T_g	35000 K	<65 °C Prusament PLA
PLA-012 Chapuis2025	k_i / branch 1	6.228 MPa	65 °C Prusament PLA
PLA-013 Chapuis2025	τ_i / branch 1	1.176×10^{-14} s	65 °C Prusament PLA
PLA-014 Chapuis2025	k_i / branch 2	1.806 MPa	65 °C Prusament PLA
PLA-015 Chapuis2025	τ_i / branch 2	1.228×10^{-14} s	65 °C Prusament PLA
PLA-016 Chapuis2025	k_i / branch 3	7.849 MPa	65 °C Prusament PLA
PLA-017 Chapuis2025	τ_i / branch 3	2.878×10^{-14} s	65 °C Prusament PLA
PLA-018 Chapuis2025	k_i / branch 4	12.751 MPa	65 °C Prusament PLA
PLA-019 Chapuis2025	τ_i / branch 4	1.701×10^{-12} s	65 °C Prusament PLA
PLA-020 Chapuis2025	k_i / branch 5	20.857 MPa	65 °C Prusament PLA
PLA-021 Chapuis2025	τ_i / branch 5	9.178×10^{-12} s	65 °C Prusament PLA
PLA-022 Chapuis2025	k_i / branch 6	31.104 MPa	65 °C Prusament PLA
PLA-023 Chapuis2025	τ_i / branch 6	6.034×10^{-11} s	65 °C Prusament PLA
PLA-024 Chapuis2025	k_i / branch 7	45.566 MPa	65 °C Prusament PLA

ID / source	Property / component	Value / unit	Temperature / formulation
PLA-025 Chapuis2025	τ_i / branch 7	6.187×10^{-9} s	65 °C Prusament PLA
PLA-026 Chapuis2025	k_i / branch 8	64.036 MPa	65 °C Prusament PLA
PLA-027 Chapuis2025	τ_i / branch 8	1.102×10^{-7} s	65 °C Prusament PLA
PLA-028 Chapuis2025	k_i / branch 9	92.484 MPa	65 °C Prusament PLA
PLA-029 Chapuis2025	τ_i / branch 9	1.047×10^{-6} s	65 °C Prusament PLA
PLA-030 Chapuis2025	k_i / branch 10	135.916 MPa	65 °C Prusament PLA
PLA-031 Chapuis2025	τ_i / branch 10	6.903×10^{-6} s	65 °C Prusament PLA
PLA-032 Chapuis2025	k_i / branch 11	113.488 MPa	65 °C Prusament PLA
PLA-033 Chapuis2025	τ_i / branch 11	2.582×10^{-5} s	65 °C Prusament PLA
PLA-034 Chapuis2025	k_i / branch 12	128.266 MPa	65 °C Prusament PLA
PLA-035 Chapuis2025	τ_i / branch 12	5.363×10^{-5} s	65 °C Prusament PLA
PLA-036 Chapuis2025	k_i / branch 13	173.47 MPa	65 °C Prusament PLA
PLA-037 Chapuis2025	τ_i / branch 13	0.0001838 s	65 °C Prusament PLA
PLA-038 Chapuis2025	k_i / branch 14	93.393 MPa	65 °C Prusament PLA
PLA-039 Chapuis2025	τ_i / branch 14	0.0004386 s	65 °C Prusament PLA
PLA-040 Chapuis2025	k_i / branch 15	163.804 MPa	65 °C Prusament PLA
PLA-041 Chapuis2025	τ_i / branch 15	0.0007577 s	65 °C Prusament PLA
PLA-042 Chapuis2025	k_i / branch 16	134.311 MPa	65 °C Prusament PLA

ID / source	Property / component	Value / unit	Temperature / formulation
PLA-043 Chapuis2025	τ_i / branch 16	0.001795 s	65 °C Prusament PLA
PLA-044 Chapuis2025	k_i / branch 17	119.266 MPa	65 °C Prusament PLA
PLA-045 Chapuis2025	τ_i / branch 17	0.002292 s	65 °C Prusament PLA
PLA-046 Chapuis2025	k_i / branch 18	100.596 MPa	65 °C Prusament PLA
PLA-047 Chapuis2025	τ_i / branch 18	0.006576 s	65 °C Prusament PLA
PLA-048 Chapuis2025	k_i / branch 19	80.534 MPa	65 °C Prusament PLA
PLA-049 Chapuis2025	τ_i / branch 19	0.00992 s	65 °C Prusament PLA
PLA-050 Chapuis2025	k_i / branch 20	61.624 MPa	65 °C Prusament PLA
PLA-051 Chapuis2025	τ_i / branch 20	0.0329 s	65 °C Prusament PLA
PLA-052 Chapuis2025	k_i / branch 21	44.508 MPa	65 °C Prusament PLA
PLA-053 Chapuis2025	τ_i / branch 21	0.05374 s	65 °C Prusament PLA
PLA-054 Chapuis2025	k_i / branch 22	30.529 MPa	65 °C Prusament PLA
PLA-055 Chapuis2025	τ_i / branch 22	0.06586 s	65 °C Prusament PLA
PLA-056 Chapuis2025	k_i / branch 23	18.61 MPa	65 °C Prusament PLA
PLA-057 Chapuis2025	τ_i / branch 23	35.03 s	65 °C Prusament PLA
PLA-058 Chapuis2025	ε_{11}^{AM} / pattern-search identification	0.05618	75 °C Prusament PLA
PLA-059 Chapuis2025	ε_{11}^{AM} / Timoshenko-bilayer identification	0.05605	75 °C Prusament PLA
PLA-060 Chapuis2025	ε_{11}^{AM} / pattern-search identification	0.04206	75 °C Prusament PLA

ID / source	Property / component	Value / unit	Temperature / formulation
PLA-061 Chapuis2025	ε_{11}^{AM} / Timoshenko-bilayer identification	0.04199	75 °C Prusament PLA
PLA-062 Chapuis2025	ε_{11}^{AM} / pattern-search identification	0.1035	90 °C Prusament PLA
PLA-063 Chapuis2025	ε_{11}^{AM} / Timoshenko-bilayer identification	0.1029	90 °C Prusament PLA
PLA-064 Chapuis2025	ε_{11}^{AM} / pattern-search identification	0.07957	90 °C Prusament PLA
PLA-065 Chapuis2025	ε_{11}^{AM} / Timoshenko-bilayer identification	0.07926	90 °C Prusament PLA
PLA-066 Luberto2024	ρ / isotropic	1240 kg m^{-3}	20 °C Unidentified PLA filament
PLA-067 Luberto2024	c_p / isotropic	1800 J kg^{-1} K^{-1}	20 °C Unidentified PLA filament
PLA-068 Luberto2024	k / isotropic	0.13 W m^{-1} K^{-1}	20 °C Unidentified PLA filament
PLA-069 Trofimov2022	E / isotropic model table	1860 MPa	25 °C Raise3D Premium PLA with transferred functions
PLA-070 Trofimov2022	E / isotropic model table	1727 MPa	40 °C Raise3D Premium PLA with transferred functions
PLA-071 Trofimov2022	E / isotropic model table	1603 MPa	50 °C Raise3D Premium PLA with transferred functions
PLA-072 Trofimov2022	E / isotropic model table	1000 MPa	60 °C Raise3D Premium PLA with transferred functions
PLA-073 Trofimov2022	E / isotropic model table	300 MPa	70 °C Raise3D Premium PLA with transferred functions
PLA-074 Trofimov2022	E / isotropic model table	50 MPa	80 °C Raise3D Premium PLA with transferred functions
PLA-075 Trofimov2022	E / isotropic model table	10 MPa	90 °C Raise3D Premium PLA with transferred functions
PLA-076 Trofimov2022	E / isotropic model table	1 MPa	100 °C Raise3D Premium PLA with transferred functions

ID / source	Property / component	Value / unit	Temperature / formulation
PLA-077 Trofimov2022	E / isotropic model table	1 MPa	150 °C Raise3D Premium PLA with transferred functions
PLA-078 Trofimov2022	α / isotropic low-temperature table value	79 μm m^{-1} K^{-1}	25–90 °C Raise3D Premium PLA with transferred functions
PLA-079 Trofimov2022	α / isotropic high-temperature table value	147 μm m^{-1} K^{-1}	100–150 °C Raise3D Premium PLA with transferred functions
PLA-080 Trofimov2022	k / isotropic low-temperature table value	0.11 W m^{-1} K^{-1}	25–90 °C Raise3D Premium PLA with transferred functions
PLA-081 Trofimov2022	k / isotropic high-temperature table value	0.19 W m^{-1} K^{-1}	100–150 °C Raise3D Premium PLA with transferred functions
PLA-082 Trofimov2022	c_p / isotropic low-temperature table value	1590 J kg^{-1} K^{-1}	25–90 °C Raise3D Premium PLA with transferred functions
PLA-083 Trofimov2022	c_p / isotropic high-temperature table value	1950 J kg^{-1} K^{-1}	100–150 °C Raise3D Premium PLA with transferred functions
PLA-084 Trofimov2022	v / isotropic	0.36	25–150 °C Raise3D Premium PLA with transferred functions
PLA-085 Trofimov2022	ρ / isotropic	1250 kg m^{-3}	25–150 °C Raise3D Premium PLA with transferred functions
PLA-086 Li2024	E_1 / road direction 1	2669 MPa	not reported FormFutura Premium PLA
PLA-087 Li2024	E_2 / transverse in-plane 2	2583 MPa	not reported FormFutura Premium PLA
PLA-088 Li2024	E_3 / build direction 3	2208 MPa	not reported FormFutura Premium PLA
PLA-089 Li2024	v_{12} / 12	0.43	not reported FormFutura Premium PLA
PLA-090 Li2024	v_{13} / 13	0.37	not reported FormFutura Premium PLA
PLA-091 Li2024	G_{12} / 12	919 MPa	not reported FormFutura Premium PLA

ID / source	Property / component	Value / unit	Temperature / formulation
PLA-092 Li2024	G_{31} / 31	844 MPa	not reported FormFutura Premium PLA
PLA-093 Relaxation2022	E / 0°	3045 MPa	not reported SUNLU PLA Plus
PLA-094 Relaxation2022	E / 45°	2914 MPa	not reported SUNLU PLA Plus
PLA-095 Relaxation2022	E / 90°	2932 MPa	not reported SUNLU PLA Plus
PLA-096 Relaxation2022	$\Delta E/E_0$ / 0°/45°/90° summary	≈11–13 %	not reported SUNLU PLA Plus

S5. Geometry, discretization and run matrix

The intended coupon is 60 mm × 10 mm × 4 mm, with two 70 mm × 20 mm × 5 mm plates. The reference-state candidate total clearances are 0, 0.01, 0.02, 0.04 and 0.08 mm. These are design choices. No production geometry mesh, heat-transfer boundary history or released-state protocol is approved.

The complete accepted verification matrix comprises one plane-wall thermal run, six structural/contact cases and 28 unique convergence runs. The 43 convergence attempt directories include 15 excluded attempts; they are not independent experiments. The table below identifies every accepted convergence run. Complete case parameters, mesh counts, time histories, contact controls and solver messages are preserved in each case.json, input.dat and mapdl.out.

Table S12. Complete accepted convergence case matrix

Run directory beneath simulation/convergence	Scope
stage10 contact augmented fkn 0p1 a03	Reference discretization / contact study; no production prediction
stage10 contact augmented fkn 10 a03	Reference discretization / contact study; no production prediction
stage10 contact augmented mu 0p1 a03	Reference discretization / contact study; no production prediction
stage10 contact augmented mu 0p3 a03	Reference discretization / contact study; no production prediction
stage10 contact mesh fine a03	Reference discretization / contact study; no production prediction
stage10 contact mesh lagrange coarse a03	Reference discretization / contact study; no production prediction
stage10 contact mesh lagrange extra fine a03	Reference discretization / contact study; no production prediction

Run directory beneath simulation/convergence	Scope
stage10 contact mesh lagrange fine a03	Reference discretization / contact study; no production prediction
stage10 contact mesh lagrange medium a03	Reference discretization / contact study; no production prediction
stage10 contact normal lagrange a03	Reference discretization / contact study; no production prediction
stage10 contact penalty fkn 0p1 a03	Reference discretization / contact study; no production prediction
stage10 contact penalty fkn 10 a03	Reference discretization / contact study; no production prediction
stage10 contact penalty fkn 1 a03	Reference discretization / contact study; no production prediction
stage10 structural mesh coarse a03	Reference discretization / contact study; no production prediction
stage10 structural mesh extra fine a01	Reference discretization / contact study; no production prediction
stage10 structural mesh fine a01	Reference discretization / contact study; no production prediction
stage10 structural mesh medium a01	Reference discretization / contact study; no production prediction
stage10 structural mesh ultra fine a01	Reference discretization / contact study; no production prediction
stage10 structural time coarse a01	Reference discretization / contact study; no production prediction
stage10 structural time extra fine a01	Reference discretization / contact study; no production prediction
stage10 structural time medium a01	Reference discretization / contact study; no production prediction
stage10 thermal mesh coarse a01	Reference discretization / contact study; no production prediction
stage10 thermal mesh extra fine a01	Reference discretization / contact study; no production prediction
stage10 thermal mesh fine a01	Reference discretization / contact study; no production prediction
stage10 thermal mesh medium a01	Reference discretization / contact study; no production prediction

Run directory beneath simulation/convergence	Scope
stage10 thermal time coarse a01	Reference discretization / contact study; no production prediction
stage10 thermal time extra fine a01	Reference discretization / contact study; no production prediction
stage10 thermal time medium a01	Reference discretization / contact study; no production prediction

S6. Validation and unavailable analyses

The validation registry contains 45 literature observations: 18 reserved source-specific means, 24 excluded context observations and three quarantined secondary maxima. No prediction or validation error is filled. The six temperature levels in Table S4 describe a separate published Ultimaker experiment and are not the proposed Prusament production matrix. Source material, cooling and thermal-contact closure remain insufficient for an independent solve.

The production design and case manifest, all-cases results, Pareto and confirmation tables contain zero result rows. There are no surrogate diagnostics, global sensitivity indices, material probability distributions, uncertainty intervals, Pareto solutions or confirmation runs. No additional production contours can be provided. This release does not claim completion of these studies.

S7. Historical reproduction record

Tables S1–S12 retain the earlier accepted calculations and material inventory. Their historical build environment, hashes and quality checks remain in the Stage 20 records. Those records describe the earlier package; the current master build, checks and environment are specified in Section M3 and reproducibility/MASTER_README.md. Every archived attempt remains immutable. Neither reference-case convergence nor the current export quality establishes physical validation.

M1. New calculations and numerical verification

The source material-point check is deliberately uniaxial and print-conditioned. Its 23–75 °C history is prescribed and differs from the original spring experiment. The spring involved two PLA formulations; the reserved spring means and newer Prusament dog-bone observations are not calibration inputs and have no prediction column populated. The current main paper and repository decision documents explain why no physical validation accuracy is claimed.

The non-isothermal ANSYS test uses a source-derived 23-arm spectrum, full Newton iteration and a temperature-shift table. The original 0.0001 MPa + 0.002|reference stress| gate is retained. Coarse solver and accuracy failures are listed below. The reference discretization is independently halved. All accepted/rejected input, command, source snapshot, log and scalar export hashes are retained in each attempt directory. Binary result files remain local optional evidence; their manifests distinguish their absence from a changed scalar export.

Table S13. Every new solver attempt and disposition

Attempt	Disposition	Reason
coarse a01	Excluded	Python FileNotFoundError [WinError 3]: scratch parent did not exist before solver launch. No solver output.
coarse a02	Excluded	Solver returned nonzero
coarse a03	Excluded	Accuracy or wrong-normal check failed
confirm a01	Accepted	Numerical checks passed
fine a01	Accepted	Numerical checks passed
medium a01	Excluded	Accuracy or wrong-normal check failed
table confirm a01	Accepted	Numerical checks passed
coexpanding inside spacers a01	Excluded	Accuracy or wrong-normal check failed
coexpanding inside spacers a02	Accepted	Numerical checks passed
fixed inside spacing a01	Excluded	Accuracy or wrong-normal check failed
fixed inside spacing a02	Accepted	Numerical checks passed
fixed outer faces a01	Excluded	Accuracy or wrong-normal check failed
fixed outer faces a02	Accepted	Numerical checks passed

Table S14. Final non-isothermal sample comparisons; stresses in MPa

Time (s)	T (°C)	ANSYS σ	Reference σ	Absolute error
1	24	−0.1149099	−0.1149099	5.28719×10^{-9}
10	33	−1.114813	−1.114815	1.476839×10^{-6}
20	43	−2.076544	−2.076547	2.217426×10^{-6}
30	53	−1.684159	−1.68418	2.094258×10^{-5}
40	63	−0.08241639	−0.08252278	0.0001063963
42	65	−0.08191326	−0.08199766	8.439955×10^{-5}
45	68	−0.07270559	−0.07274731	4.171689×10^{-5}
50	73	−0.03626523	−0.03626813	2.898231×10^{-6}
55	78	−0.03964798	−0.03965075	2.772462×10^{-6}
57	80	−0.04107951	−0.04108327	3.76578×10^{-6}
60	80	−0.04107785	−0.04107785	5.238453×10^{-10}
70	80	−0.04107785	−0.04107785	5.238526×10^{-10}
90	80	−0.04107785	−0.04107785	5.238552×10^{-10}
117	80	−0.04107785	−0.04107785	5.238552×10^{-10}
120	77	−0.03889468	−0.03889266	2.016738×10^{-6}
125	72	−0.03499196	−0.03498697	4.991189×10^{-6}
130	67	−0.02878648	−0.02873358	5.290454×10^{-5}
132	65	−0.02488581	−0.02480134	8.447737×10^{-5}
140	57	0.04362486	0.04368606	6.120436×10^{-5}
150	47	0.644797	0.6448054	8.437489×10^{-6}
160	37	1.674599	1.674601	1.577862×10^{-6}
170	27	2.792315	2.792312	3.190741×10^{-6}
174	23	3.249814	3.24981	3.3466×10^{-6}
180	23	3.248701	3.248698	3.123272×10^{-6}
200	23	3.247546	3.247543	3.246339×10^{-6}
234	23	3.247008	3.247004	3.310843×10^{-6}

Table S15. Programmed-road refinement; dimensionless mechanical and total strains

Substeps	Phase	Mechanical strain	Total strain	Net stress (MPa)
8	heat	−0.05322308	−0.04968708	3.6107×10^{-17}
8	hold	−0.056175	−0.052639	4.22865×10^{-26}
8	cool	−0.056175	−0.056175	-8.24466×10^{-30}
32	heat	−0.0556399	−0.0521039	7.90716×10^{-19}
32	hold	−0.056175	−0.052639	8.34342×10^{-39}
32	cool	−0.056175	−0.056175	1.50348×10^{-40}
128	heat	−0.05610269	−0.05256669	2.54963×10^{-20}
128	hold	−0.056175	−0.052639	2.79107×10^{-49}
128	cool	−0.056175	−0.056175	9.84877×10^{-51}
512	heat	−0.05613575	−0.05259975	-1.92181×10^{-20}
512	hold	−0.056175	−0.052639	3.57585×10^{-50}
512	cool	−0.056175	−0.056175	9.46847×10^{-53}

The following stop cases use half-thickness symmetry, rigid plane motion, a constant equilibrium spring, prescribed uniform temperature and no nonthermal recovery. Force is the full 600 mm^2 face resultant. Reported gap is the one-sided node-to-stop separation; total clearance is twice that value. These are benchmarks and not pressure distributions.

Table S16. Three-dimensional stop outputs; state 1 open, state 2 closed

Mount	Load	T (°C)	Gap (mm)	Force (N)	State
Inner fixed	1	30	0.009048	−0	1
Inner fixed	2	45	0.007008	−0	1
Inner fixed	3	60	0.004968	−0	1
Inner fixed	4	70	0.003608	−0	1
Inner fixed	5	75	0.002928	−0	1
Inner fixed	6	80	0.002248	−0	1
Inner fixed	7	75	0.002928	−0	1
Inner fixed	8	70	0.003608	−0	1
Inner fixed	9	60	0.004968	−0	1

Mount	Load	T (°C)	Gap (mm)	Force (N)	State
Inner fixed	10	45	0.007008	−0	1
Inner fixed	11	30	0.009048	−0	1
Inner fixed	12	23	0.01	−0	1
Inner spacers	1	30	0.009273109	−0	1
Inner spacers	2	45	0.007715486	−0	1
Inner spacers	3	60	0.006157863	−0	1
Inner spacers	4	70	0.005119448	−0	1
Inner spacers	5	75	0.00460024	−0	1
Inner spacers	6	80	0.004081032	−0	1
Inner spacers	7	75	0.00460024	−0	1
Inner spacers	8	70	0.005119448	−0	1
Inner spacers	9	60	0.006157863	−0	1
Inner spacers	10	45	0.007715486	−0	1
Inner spacers	11	30	0.009273109	−0	1
Inner spacers	12	23	0.01	−0	1
Outer fixed	1	30	0.008488026	−0	1
Outer fixed	2	45	0.005248085	−0	1
Outer fixed	3	60	0.002008142	−0	1
Outer fixed	4	70	−0	0.482695	2
Outer fixed	5	75	−0	3.916386	2
Outer fixed	6	80	−0	7.350077	2
Outer fixed	7	75	−0	3.916386	2
Outer fixed	8	70	−0	0.482695	2
Outer fixed	9	60	0.002008142	−0	1
Outer fixed	10	45	0.005248085	−0	1
Outer fixed	11	30	0.008488026	−0	1
Outer fixed	12	23	0.01	−0	1

M2. Complete deterministic clearance screen

The critical signed free thickness strain comes from geometry and thermal expansion; it is not an observed or fitted recovery strain. I = fixed inner spacing; S = coexpanding inner spacers; O = fixed outer faces. All calculations use h_0 = 4 mm and plate thickness 5 mm. The mean steel expansion is treated as constant within the manufacturer interval and rebased to 23 °C. No parameter probability or contact-pressure prediction is introduced.

Table S17. All 90 kinematic scenarios

T (°C)	g_0 (mm)	Mount	ΔH (mm)	e critical (%)
60	0	I	0	−0.2516
60	0.01	I	0	−0.0016
60	0.02	I	0	0.2484
60	0.04	I	0	0.7484
60	0.08	I	0	1.7484
60	0	S	0.002367886	−0.1924028
60	0.01	S	0.002373806	0.05774515
60	0.02	S	0.002379726	0.3078931
60	0.04	S	0.002391565	0.8081891
60	0.08	S	0.002415244	1.808781
60	0	O	−0.005919716	−0.3995929
60	0.01	O	−0.005919716	−0.1495929
60	0.02	O	−0.005919716	0.1004071
60	0.04	O	−0.005919716	0.6004071
60	0.08	O	−0.005919716	1.600407
65	0	I	0	−0.2856
65	0.01	I	0	−0.0356
65	0.02	I	0	0.2144
65	0.04	I	0	0.7144
65	0.08	I	0	1.7144
65	0	S	0.002687871	−0.2184032

T (°C)	g_0 (mm)	Mount	ΔH (mm)	e critical (%)
65	0.01	S	0.002694591	0.03176477
65	0.02	S	0.00270131	0.2819328
65	0.04	S	0.00271475	0.7822687
65	0.08	S	0.002741628	1.782941
65	0	O	−0.006719677	−0.4535919
65	0.01	O	−0.006719677	−0.2035919
65	0.02	O	−0.006719677	0.04640806
65	0.04	O	−0.006719677	0.5464081
65	0.08	O	−0.006719677	1.546408
70	0	I	0	−0.3196
70	0.01	I	0	−0.0696
70	0.02	I	0	0.1804
70	0.04	I	0	0.6804
70	0.08	I	0	1.6804
70	0	S	0.003007856	−0.2444036
70	0.01	S	0.003015375	0.005784382
70	0.02	S	0.003022895	0.2559724
70	0.04	S	0.003037934	0.7563484
70	0.08	S	0.003068013	1.7571
70	0	O	−0.007519639	−0.507591
70	0.01	O	−0.007519639	−0.257591
70	0.02	O	−0.007519639	−0.007590976
70	0.04	O	−0.007519639	0.492409
70	0.08	O	−0.007519639	1.492409
75	0	I	0	−0.3536
75	0.01	I	0	−0.1036
75	0.02	I	0	0.1464

T (°C)	g_0 (mm)	Mount	ΔH (mm)	e critical (%)
75	0.04	I	0	0.6464
75	0.08	I	0	1.6464
75	0	S	0.00332784	−0.270404
75	0.01	S	0.00333616	−0.020196
75	0.02	S	0.003344479	0.230012
75	0.04	S	0.003361119	0.730428
75	0.08	S	0.003394397	1.73126
75	0	O	−0.008319601	−0.56159
75	0.01	O	−0.008319601	−0.31159
75	0.02	O	−0.008319601	−0.06159002
75	0.04	O	−0.008319601	0.43841
75	0.08	O	−0.008319601	1.43841
80	0	I	0	−0.3876
80	0.01	I	0	−0.1376
80	0.02	I	0	0.1124
80	0.04	I	0	0.6124
80	0.08	I	0	1.6124
80	0	S	0.003647825	−0.2964044
80	0.01	S	0.003656944	−0.04617639
80	0.02	S	0.003666064	0.2040516
80	0.04	S	0.003684303	0.7045076
80	0.08	S	0.003720781	1.70542
80	0	O	−0.009119562	−0.6155891
80	0.01	O	−0.009119562	−0.3655891
80	0.02	O	−0.009119562	−0.1155891
80	0.04	O	−0.009119562	0.3844109
80	0.08	O	−0.009119562	1.384411

T (°C)	g_0 (mm)	Mount	ΔH (mm)	e critical (%)
85	0	I	0	−0.4216
85	0.01	I	0	−0.1716
85	0.02	I	0	0.0784
85	0.04	I	0	0.5784
85	0.08	I	0	1.5784
85	0	S	0.00396781	−0.3224048
85	0.01	S	0.003977729	−0.07215677
85	0.02	S	0.003987649	0.1780912
85	0.04	S	0.004007488	0.6785872
85	0.08	S	0.004047166	1.679579
85	0	O	−0.009919524	−0.6695881
85	0.01	O	−0.009919524	−0.4195881
85	0.02	O	−0.009919524	−0.1695881
85	0.04	O	−0.009919524	0.3304119
85	0.08	O	−0.009919524	1.330412

M3. Reproduction, unresolved results and data separation

Run the Python unit suite and scripts/screen_fixture_mechanism.py, scripts/check_programmed_recovery.py, scripts/summarize_master_nonisothermal.py --audit-reference, and scripts/check_master.py from the repository root. The first two perform numerical calculations; the third recomputes comparison summaries from immutable exports. No ANSYS license is needed for these checks. scripts/plot_master.py draws the figures; scripts/build_master.py and scripts/build_master_supplement.py regenerate the documents from their maintained sources. The exact tested dependency versions and font/Office requirements are documented in reproducibility/MASTER_README.md. The main PDF is a native Word export of the editable manuscript; this supplement is typeset separately.

To repeat a solver case, use compatible Ansys Student 2026 R1/MAPDL26.1 with a valid license and a new empty working directory; run its archived input.dat with the recorded command adapted to the new path. Never reuse an archived attempt directory. The master runners accept a new attempt identifier and preserve accepted and failed evidence. They do not assemble a validated production coupon. One SOLID185 plus one CONTA178 in the stop test is far below the documented Student node limit but is not a converged production surface mesh.

No production design is approved. All production, sensitivity, uncertainty, surrogate, optimization and confirmation registries remain empty. These absences are not zero errors, zero warpage, an empty feasible set or a failed physical validation score. Source-specific reserved observations are in

validation/master_spring_targets.csv and validation/master_suder_targets.csv; their original cohorts and limitations are retained. Earlier Ultimaker means remain in the legacy register. No dataset is both fitted and independently validated here.

The code-generated figures contain no AI-generated result imagery. Raw scalar evidence, equation/source links and all final artifacts are hashed. The prior full numerical tables are retained above; the duplicate thermal case-label ambiguity in S10 is corrected by keeping mesh/time in the names. The following legacy bibliography permits independent interpretation of all prior numeric citations. New citations in the main article use its own bibliography.

M4. Legacy reference key for Tables S1–S12

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